

Cross-Session Tool-Trajectory Detection for LLM Agents: A Reproducible Empirical Evaluation

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ABSTRACT

Indirect prompt injection and multi-round attacks can distribute a harmful objective across otherwise ordinary tool calls. This paper evaluates a reduced trajectory-only candidate derived from a broader BehaviorGraph implementation. The candidate was selected using development evidence (R1-R3), frozen before two independent API confirmation runs (R4-R5), and calibrated without test-set threshold selection. Across 480 attack groups and 400 benign test groups, the candidate detects 139 attack attempts (29.0%), recalls 137 of 320 successful objectives (42.8%), and raises one benign alert (0.25%). Exact total-tool-call-count matching reduces detection to 22.2%-25.7%. Transition/frequency deviations provide the measured signal; removing the cross-session component changes detection by less than one percentage point. Removing cumulative scoring reaches 95%-100% in a post-confirmation exploratory ablation and therefore requires preregistered replication. The results are specific to one model family and a banking-style sandbox and do not support production-readiness or cross-model claims.

Index Terms - LLM agents, tool trajectories, cross-session detection, prompt injection, reproducible evaluation.

1. INTRODUCTION

Tool-using agents transform model outputs into actions such as reading messages, exporting data, and executing transactions. A harmful goal may be fragmented across sessions, so single-turn content detectors and trajectory-only monitors address different failure modes. This study asks a narrower empirical question: after candidate selection is separated from confirmation, what detection-false-alarm trade-off is obtained from a trajectory score?

We distinguish the complete BehaviorGraph framework (Full-BG) from the pre-registered trajectory-only confirmation candidate (TO-CF). Full-BG contains graph-structure, parameter, combination, transition/frequency, and cumulative components. TO-CF disables the first three optional components and is the only configuration independently confirmed in R4/R5. This distinction prevents a graph-novelty claim from being inferred from results that do not measure the full feature set.

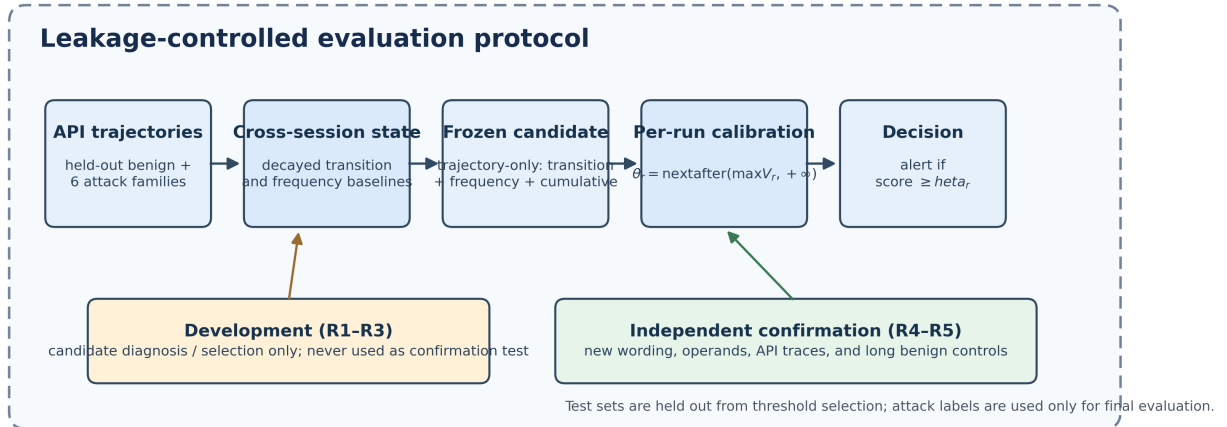


Fig. 1. Leakage-controlled protocol. R1-R3 are development evidence only; TO-CF is frozen before R4-R5, which use new wording, operands, API trajectories, and long normal controls.

2. CONFIRMED CANDIDATE AND SCORE PROTOCOL

For each ordered group g , the evaluator retains the transition/frequency baseline across sessions and computes a cumulative EWMA score $q(c)$ after every call c . The group statistic is $S(g) = \max$ over calls c of $q(c)$. For run r , the threshold is $\theta_{eta_r} = \text{nextafter}(\text{maximum validation-group } S(g), +\infty)$, where the validation set contains 100 benign groups (50 short and 50 long). A group is alerted iff $S(g) \geq \theta_{eta_r}$. The evaluator applies this external score protocol and does not use the default online `get_decision` threshold.

The canonical thresholds are 5.0040541339 (R4) and 4.9817326042 (R5). P95 is retained only in legacy/development scripts and is not the R4/R5 rule.

3. EXPERIMENTAL DESIGN

Each confirmation run contains 240 attack groups, 200 benign test groups, 100 validation groups, and 200 calibration groups. There are 40 attack groups per family across six families: delayed trigger, indirect injection, persistent memory, privilege escalation, tool-chain exfiltration, and transaction cover. The successful-objective denominator is 160 per run because delayed-trigger and persistent-memory attempts did not produce a successful objective under the protocol. Bootstrap intervals resample groups, not individual calls.

TABLE I. Configuration boundary. Only TO-CF is an independent confirmation claim.

Item	Full-BG framework	TO-CF confirmation
Structure score	enabled by default	disabled
Parameter rules	enabled by default	disabled
Tool-combination rules	enabled by default	disabled
Transition/frequency	available	enabled
Cumulative score	available	enabled
Independent confirmation	not run	R4 and R5

4. INDEPENDENT CONFIRMATION

TABLE II. TO-CF confirmation results. Intervals and full denominators are in the machine-readable audit.

Metric	R4	R5
Threshold	5.004054	4.981733
All-attempt detection	68/240 = 28.3%	71/240 = 29.6%
Successful recall	67/160 = 41.9%	70/160 = 43.8%
Benign FPR	0/200 = 0.0%	1/200 = 0.5%
Exact length matched	26/117 = 22.2%	29/113 = 25.7%

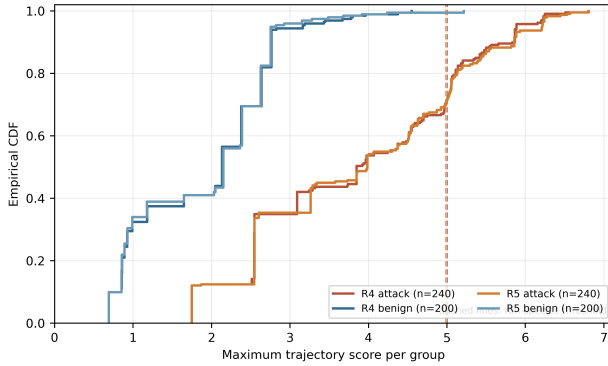


Fig. 2. Empirical score distributions for attack and benign groups. Dashed lines are the per-run validation-maximum thresholds; the overlap explains limited recall at low false-positive rates.

The two runs are numerically similar, but two repetitions do not establish broad stability. We therefore report them separately and use the aggregate only as a descriptive macro summary: 29.0% all-attempt detection, 42.8% successful-objective recall, and 0.25% FPR.

5. ATTACK HETEROGENEITY AND DIAGNOSTICS

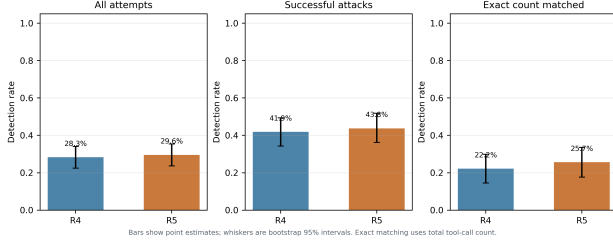


Fig. 3. Primary TO-CF metrics across R4/R5. Error bars are bootstrap 95% intervals; exact matching is by total tool-call count.

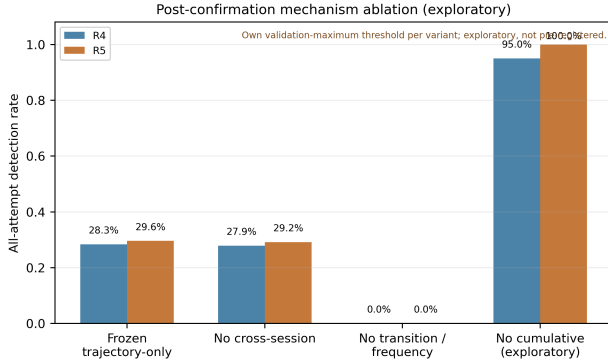


Fig. 4. Post-confirmation exploratory ablation. Each variant uses its own validation-maximum calibration; the no-cumulative result is not a confirmed method claim.

TABLE III. All-attempt detection by attack family. Successful-objective recall has a separate denominator of 160 per run.

Attack family	R4	R5
Delayed trigger	0/40 = 0%	0/40 = 0%
Indirect injection	0/40 = 0%	0/40 = 0%
Persistent memory	1/40 = 2.5%	1/40 = 2.5%
Privilege escalation	35/40 = 87.5%	37/40 = 92.5%
Tool-chain exfiltration	16/40 = 40%	17/40 = 42.5%
Transaction cover	16/40 = 40%	16/40 = 40%

The mechanism result is deliberately limited. No cross-session changes detection from 28.3% to 27.9% in R4 and from 29.6% to 29.2% in R5. Removing transition/frequency signals yields 0% in both runs. Removing cumulative scoring yields 95.0% and 100.0%, but this choice was made after observing R4/R5 and must be tested in a new preregistered R6/R7 study.

6. FAILURE ANALYSIS AND BASELINE CONTEXT

The primary candidate misses 183 successful objectives across R4/R5. The misses are concentrated in indirect injection (80, 44%), tool-chain exfiltration (47, 26%), transaction cover (48, 26%), and privilege escalation (8, 4%). These cases show that a harmful objective can be completed through calls that remain close to normal tool-use patterns; trajectory-only evidence cannot recover information absent from the calls.

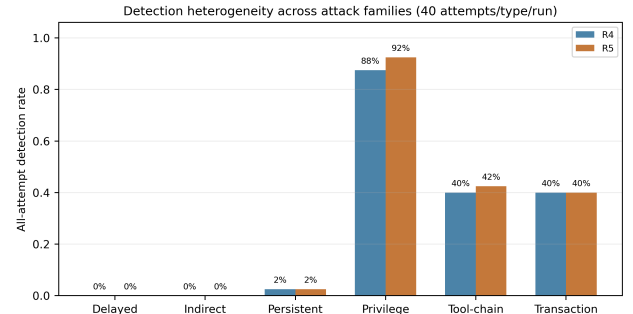


Fig. 5. All-attempt detection heterogeneity across the six attack families (40 groups per family per run). This is an attack-type breakdown, not a missed-case-only chart.

AgentShield was evaluated in a separate defended execution because its honeytools and honeytokens change the trajectory. It achieved 67.1%/66.7% all-attempt detection, 100% successful-objective recall, and 0.5%/0.0% FPR in R4/R5. These are descriptive reference measurements, not paired replays or a causal superiority test.

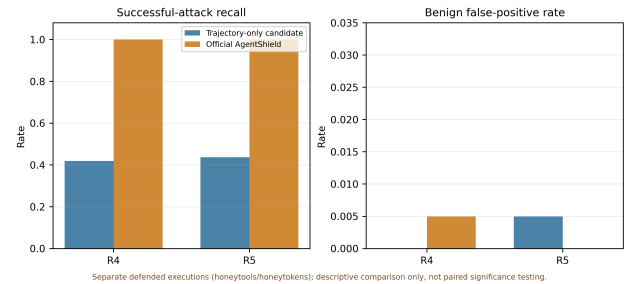


Fig. 6. Separate-execution comparison with the official AgentShield adapter. Honeytool/honeytoken intervention changes the environment; no paired significance claim is made.

7. LIMITATIONS AND CONCLUSION

The study evaluates one API-accessible model family and a banking-style sandbox. It does not establish cross-model transfer, natural-traffic coverage, adaptive-attack robustness, or deployment latency. The FPR estimate is based on 400 independent benign test groups. Exact length matching reduces but does not eliminate confounding. Full-BG was not independently confirmed, and no-cumulative remains post-hoc.

The defensible conclusion is narrow: a locked transition/frequency trajectory score detects a measurable but limited fraction of multi-step attacks at a low observed false-positive rate, while the measured incremental value of the cross-session graph component is small in this benchmark. The next method claim requires preregistered R6/R7, a hybrid content-plus-trajectory detector, or additional model/domain confirmation.

References

- [1] Y. H. Rassul and T. A. Rashid, "AgentShield: Deception-based Compromise Detection for Tool-using LLM Agents," arXiv:2605.11026, 2026.
- [2] A. Mehta et al., "FragBench: Cross-Session Attacks Hidden in Benign-Looking Fragments," arXiv:2605.11029, 2026.